Day 2 Lecture Notes: Vectors, Matrices, and Systems

Mathematical Modeling & Computational Projects Camp

Morning Structure (10:00–12:30)

- 10:00–10:30: Lecture Block 1
- 10:35–11:05: Problem Solving Session 1 (30 minutes)
- 11:10–11:40: Lecture Block 2
- 11:40–12:10: Problem Solving Session 2 (shorter) + 12:10–12:30 synthesis/review

Lecture Block 1 (10:00–10:30): Vectors and Matrix Language

Vectors

A vector is an ordered list, such as

$$\mathbf{v} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}.$$

Operations:

- Add vectors entry by entry.
- Multiply by scalar entry by entry.

  Instructor prompt: "If a vector has two entries, what could they represent in real life?" Example interpretations:

- $\begin{bmatrix} \text{hours of study} \\ \text{hours of sleep} \end{bmatrix}$
- $\begin{bmatrix} \text{student tickets} \\ \text{adult tickets} \end{bmatrix}$

  Key idea: a vector stores related quantities in one object.

Matrices

A matrix is a rectangular array of numbers, e.g.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

Matrices can represent data and transformations.

Interpretation example:

$$A = \begin{bmatrix} 6 & 9 \\ 4 & 7 \end{bmatrix}$$

could represent prices where row 1 is one store and row 2 is another store, and columns are two item types.

Entry notation reminder: a_{ij} means row i, column j.

Dimensions

If A is $m \times n$ and B is $n \times p$, then AB is defined and has size $m \times p$.

Dimension check rule students should memorize:

- Inside numbers must match (columns of first = rows of second).
- Outside numbers become the result size.

Quick example:

$$(2 \times 3)(3 \times 1) \to (2 \times 1) \quad \text{defined,}$$

but

$$(2 \times 3)(2 \times 2) \quad \text{not defined.}$$

Problem Solving Session 1 (10:35–11:05): Practice Set A (30 minutes)

Goal: Fluency with vector operations, matrix-vector products, and dimension checks.

Suggested pacing:

- First 10 minutes: Problems 1–4.
- Next 12 minutes: Problems 5–8.
- Last 8 minutes: Problems 9–10 and discussion.

1. Compute $\begin{bmatrix} 3 \\ -1 \end{bmatrix} + \begin{bmatrix} 4 \\ 2 \end{bmatrix}$.
2. Compute $2 \begin{bmatrix} 1 \\ 5 \end{bmatrix} - \begin{bmatrix} 3 \\ 4 \end{bmatrix}$.
3. For $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$, compute $A\mathbf{x}$.
4. Is AB defined if A is 2×3 and B is 3×2? What size is AB?
5. Is BA defined for the same matrices in Problem 4? What size?
6. Interpret entry a_{21} in words.
7. Give one real meaning of a 2-entry vector in school life.
8. Give one real meaning of a 2x2 matrix.

9. Let $M = \begin{bmatrix} 1 & 4 & 2 \\ 0 & 3 & 5 \end{bmatrix}$ and $\mathbf{u} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$. Compute $M\mathbf{u}$.

10. Explain why matrix multiplication order matters by giving dimensions where AB is defined but BA is not.

Instructor checkpoint questions:

- "What does each entry of $A\mathbf{x}$ mean in context?"
- "How can you reject invalid products quickly without doing arithmetic?"

Lecture Block 2 (11:10–11:40): Systems as Models

From words to equations

Example: student tickets s, adult tickets a.

$$\begin{cases} s + a = 120 \\ 5s + 8a = 780 \end{cases}$$

First equation counts tickets; second counts revenue.

Modeling interpretation:

- Equation 1 is a quantity constraint.
- Equation 2 is a money constraint.
- The solution must satisfy both at the same time.

Worked example

Solve

$$\begin{cases} 2x + y = 11 \\ x - y = 1 \end{cases}$$

From second equation, $y = x - 1$. Substitute:

$$2x + (x - 1) = 11 \Rightarrow 3x = 12 \Rightarrow x = 4, \ y = 3.$$

Check step (always required):

$$2(4) + 3 = 11, \quad 4 - 3 = 1.$$

Both true, so the solution is valid.

Second worked example (elimination):

$$\begin{cases} 2x + 3y = 19 \\ 4x - 3y = 5 \end{cases}$$

Add equations to eliminate y:

$$6x = 24 \Rightarrow x = 4.$$

Substitute back:

$$2(4) + 3y = 19 \Rightarrow 3y = 11 \Rightarrow y = \frac{11}{3}.$$

Point to emphasize: different solving methods can reach the same correct solution.

Why this matters

Systems model situations with several conditions at once.

Vector/matrix form preview:

$$\begin{bmatrix} 1 & 1 \\ 5 & 8 \end{bmatrix} \begin{bmatrix} s \\ a \end{bmatrix} = \begin{bmatrix} 120 \\ 780 \end{bmatrix}.$$

This is the bridge to afternoon Python with `numpy.linalg.solve`.

Problem Solving Session 2 (11:40–12:10): Practice Set B (shorter)

Goal: Build systems from context and solve with interpretation.

Suggested pacing:

- First 12 minutes: Problems 1–4.
- Next 10 minutes: Problems 5–7.
- Last 8 minutes: Problems 8–9 and solution share-out.

1. Solve: $\begin{cases} x + 2y = 9 \\ 3x - y = 5 \end{cases}.$
2. Write a 2-equation system for two snack types with known total count and total cost.
3. A matrix product is not defined. What dimension check should you do first?
4. Explain in one sentence how vectors and systems are connected.
5. Ticket model: total 200 tickets, prices 6 and 10, revenue 1520. Write (do not solve) a system.
6. Solve by substitution: $\begin{cases} 2x + y = 13 \\ x - y = 2 \end{cases}.$
7. Solve by elimination: $\begin{cases} 3x + 2y = 16 \\ 3x - 2y = 8 \end{cases}.$
8. Cafeteria model: sandwiches s and salads d. Total items: 90. Prices: \$5 and \$7. Total revenue: \$570. Write and solve the system.
9. Reflection: in one sentence, describe one assumption in Problem 8 and how it could fail in reality.

Instructor move for fast finishers: ask students to convert one of their systems into matrix form $A\mathbf{x} = \mathbf{b}$.

Synthesis and Review (12:10–12:30)

- Class check of one vector problem and one system problem.
- Common errors: dimension mismatch and sign mistakes.
- Exit check: "What does each equation mean in context?"

- Exit check 2: “How do you know if a matrix product is defined before computing?”
- Exit check 3: “Why should we plug our solution back into both equations?”

Python Preview (Afternoon)

Use NumPy arrays for vectors/matrices and `numpy.linalg.solve` for systems.